…… :QPCODE Reg.No :……

Name :……

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) FOURTH SEMESTER EXAMINATION

MG4DSCPHY200: WAVE OPTICS

Time:- $1\frac{1}{2}$ **Hour** **Marks:- 50**

Part A

Short Answer Type Questions

Answer any 10 questions

Each question carries 2 marks

$(10 \times 2 = 20)$

1. Write down the complex representation of a plane wave. [K][CO:1]
2. What do you understand by the term phase velocity? [U][CO:1]
3. Write down the equation for a plane progressive wave [K][CO:1]
4. What is Dichroism? Give one example for a dichroic crystal. [U][CO:2]
5. Distinguish between plane polarized light and circularly polarized light. [U][CO:2]
6. What is meant by interference of light? [U][CO:3]

7. Illustrate how phasor representation facilitates the addition of two sinusoidal functions differing in phase. [A][CO:3]

8. What are the conditions for sustained interference? [U][CO:3]

9. What will happen if a transparent thin sheet is introduced in the path of one of the interfering beams? [A][CO:3]

10. What are Newton's rings [U][CO:3]

11. Distinguish between Fresnel and Fraunhofer diffraction. [U][CO:4]

12. What is meant by missing orders in double slit Fraunhofer diffraction pattern? [U][CO:4]

13. What is meant by plane transmission grating? [A][CO:4]

14. Give two differences between a zone plate and a convex lens. [U][CO:4]

Part B

Short Essay Type Questions

Answer any 8 questions

Each question carries 5 marks

$(8 \times 5 = 40)$

15. State and explain Malu's law with neat diagram. [U][CO:2]

16. Calculate the least thickness of a calcite plate which would convert plane polarized light to circularly polarized light. Given $\mu_0 = 1.658$ and $\mu_e = 1.486$ and wavelength of the light is $589nm$. [U][CO:2]

17. Prove that when two electromagnetic plane waves of equal amplitude E_0, originating from two sources separated by a distance a and propagating at an angleθ, interfere on the xy-plane, the resulting irradiance distribution follows a cosine-squared dependence as given by the equation $I(y) = 4E_0^2 \cos^2\left[\frac{\pi}{\lambda} a \sin(\theta)\right]$. [A][CO:3]

18. A soap film surrounded by air has an index of refraction of 1.38. If a region of the film appears bright red ($\lambda = 633nm$) in normally reflected light, what is its minimum thickness there? [A][CO:3]

19. Describe the construction and working of Michelson's Interferometer [U][CO:3]

20. A screen is placed $2m$ away from a narrow slit. Find the slit width if the first minima occur $5mm$ on either side of the central maximum when plane waves of wavelength $600nm$ are incident on it? [A][CO:4]

21. What is the highest order spectrum which may be seen with a light of wavelength $500nm$ by means of a grating with 3000 lines per cm? [A][CO:4]

22. Prove that area of half period zones in a zone plate is a constant. [U][CO:4]